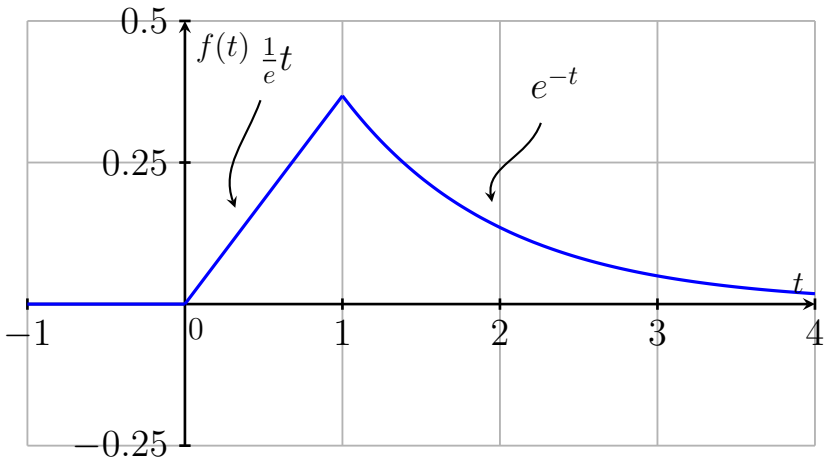
ECE 3210 hw08

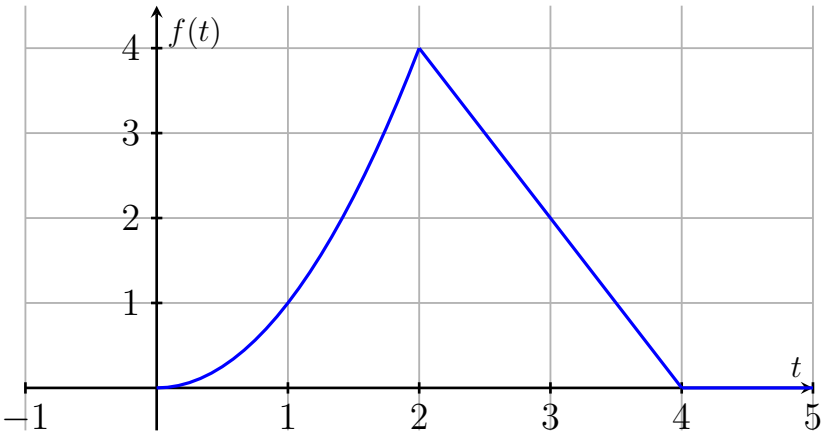
1. The following figure shows the signal $f(t)$.



   (a) Find the Laplace transform of this signal $F(s)$.

   (b) What is the region of convergence (ROC) for this Laplace transform?

2. The following figure shows the signal $f(t)$.



(a) Find the Laplace transform of this signal $F(s)$.

(b) What is the region of convergence (ROC) for this Laplace transform?

3. Consider the following Lapalace transforms $F(s)$. Find the inverse Laplace transform $f(t)$ for each case.

   (a)

$$F(s) = \frac{s+1}{s^2+6s+25}$$

   (b)

$$F(s) = \frac{5}{s^2(s+1)}$$

Answer Key

1(a) $F(s) = \frac{1}{e}\left(\frac{1}{s^2} - \frac{e^{-s}}{s^2} - \frac{e^{-s}}{s} + \frac{e^{-s}}{s+1}\right)$
1(b) The ROC is $\Re(s) > -1$.
2(a) $F(s) = \frac{2}{s^3} - \frac{2e^{-2s}}{s^3} - \frac{6e^{-2s}}{s^2} + \frac{2e^{-4s}}{s^2}$
2(b) The ROC is all values of s.
3(a) $f(t) = e^{-3t}\left(\cos(4t) - \frac{1}{2}\sin(4t)\right)u(t)$
3(b) $f(t) = (-5 + 5t + 5e^{-t})u(t)$